A scalarwise smooth counterexample to the finite-order weak–strong principle in a generalised Schwartz space

Run `fa_banach_001` — `agent_lane_11`

13 August 2026

Status. Candidate full counterexample; likely valid subject to human review. The result answers the specific generalised-Schwartz follow-up in Question 5.3.9(i) of Kruse [1].

1 The source question and the answer

For an open set $\Omega \subset \mathbb{R}^d$, a locally convex Hausdorff space E, and $k < \infty$, the source calls the implication

$$e' \circ f \in \mathcal{C}^k(\Omega) \quad (e' \in E') \qquad \Longrightarrow \qquad f \in \mathcal{C}^k(\Omega, E)$$

the full $\mathcal{C}^k$ weak–strong principle. It proves this for semi-Montel targets and asks whether the $k = 0$ result for generalised Schwartz spaces continues to hold when $0 < k < \infty$.

is Lipschitz continuous as $f \in \mathcal{C}_{loc}(\Omega, E)$. This yields that F is differentiable on $[0,1]$ almost everywhere because E is a generalised Gelfand space, implying that $(\partial^{k-1})^E f$ is differentiable on $[a,b]$ almost everywhere. Since the open set $\Omega \subset \mathbb{R}$ can be written as a countable union of disjoint open intervals I_n, $n \in \mathbb{N}$, and each I_n is a countable union of closed bounded intervals $[a_m, b_m]$, $m \in \mathbb{N}$, our statement follows from the fact that the countable union of null sets is a null set. □

To the best of our knowledge there are still some open problems for continuously partially differentiable functions of finite order.

5.3.9. Question. (i) Are there other spaces than semi-Montel spaces E for which the 'full' $\mathcal{C}^k$-weak-strong principle Theorem 5.3.6 with $k < \infty$ is true? For instance, if $k = 0$, then it is still true if E is a generalised Schwartz space by [17, 2.10 Lemma, p. 140]. Does this hold for $0 < k < \infty$ as well?

(ii) Does the 'almost' $\mathcal{C}^k$-weak-strong principle Corollary 5.3.8 also hold for $d > 1$?

(iii) For every $\varepsilon > 0$ does there exist a function $g \in \mathcal{C}^k(\mathbb{R}, E)$ such that $\lambda(\{x \in \Omega \mid f(x) \neq g(x)\}) < \varepsilon$ in Corollary 5.3.8 where λ is the one-dimensional Lebesgue measure. In the case that $E = \mathbb{R}^n$ this is true by [66, Theorem 3.1.15, p. 227].

The answer is *no for every finite* $k \geq 1$. The obstruction is exactly the distinction between precompactness in an incomplete space and compactness inside the space.

2 The target space

Let

$$E = c_{00} := \{x = (x_n)_{n \geq 1} \in \mathbb{K}^{\mathbb{N}} : x_n = 0 \text{ for all but finitely many } n\},$$

over $\mathbb{K} = \mathbb{R}$ or $\mathbb{C}$, with the topology induced by the product topology of $\mathbb{K}^{\mathbb{N}}$. Thus the seminorms

$$p_F(x) := \max_{n \in F} |x_n|, \qquad F \subset \mathbb{N} \text{ finite},$$

generate the topology.

Lemma 1. *The space E is a generalised Schwartz space, E' consists exactly of the finite linear combinations of the coordinate projections, and E is not semi-Montel.*

Proof. If $\ell \in E'$ is continuous, some basic zero-neighbourhood controlling only a finite coordinate set F is mapped by ℓ into a bounded subset of $\mathbb{K}$. Every vector vanishing on F can be multiplied by arbitrary scalars while remaining in that neighbourhood, so ℓ vanishes on it. Hence ℓ depends only on the coordinates in F. The converse is immediate.

A set $B \subset E$ is bounded precisely when

$$\sup_{x \in B} |x_n| < \infty \qquad \text{for every } n \in \mathbb{N}.$$

In particular, E' determines boundedness, as required in the source theorem. Given a zero-neighbourhood that controls finitely many coordinates, project B onto those coordinates. This bounded set has a finite net. Lifting its net points as finite-support vectors gives a finite net for B in E. Consequently every bounded set is precompact, which is the generalised Schwartz property used in the source.

Finally, set $s_N = (1, \ldots, 1, 0, \ldots)$, with the first N coordinates equal to one. The set $\{s_N : N \geq 1\}$ is bounded. Every subsequence with indices tending to infinity converges in the completion $\mathbb{K}^{\mathbb{N}}$ to $(1, 1, \ldots)$, which is not in E, and therefore has no convergent subsequence in E. Since E is metrizable, this bounded set cannot have compact closure. Thus E is not semi-Montel. $\square$

3 The counterexample

Theorem 1. *For every integer $k \geq 1$ there is a map $f_k : \mathbb{R} \to E$ such that*

(a) *$e' \circ f_k \in \mathcal{C}^\infty(\mathbb{R})$ for every $e' \in E'$;*

(b) *$f_k \in \mathcal{C}^{k-1}(\mathbb{R}, E)$;*

(c) *f_k is not k times differentiable at zero, and hence is not in $\mathcal{C}^k(\mathbb{R}, E)$.*

In particular, the full finite-order weak–strong principle fails for the generalised Schwartz space E for every $k \geq 1$.

Proof. Choose $\chi \in \mathcal{C}_c^\infty(\mathbb{R})$ such that $\chi = 1$ on a neighbourhood of zero, and define

$$f_k(t) := \big(t^k \chi(nt)\big)_{n \geq 1}. \tag{1}$$

If $t \neq 0$, compactness of $\operatorname{supp} \chi$ implies that $\chi(nt) = 0$ for every sufficiently large n; and $f_k(0) = 0$. Thus $f_k(t) \in c_{00}$ for every t.

By Lemma 1, every $e' \in E'$ uses only finitely many coordinates. Therefore $e' \circ f_k$ is a finite linear combination of the smooth scalar functions $t \mapsto t^k \chi(nt)$, proving (a).

For $0 \leq j < k$, the coordinatewise jth derivative at zero is zero. Away from zero the family in (1) is locally finite, so all derivatives are obtained coordinatewise and remain in c_{00}. The product-subspace topology is precisely coordinatewise convergence on finite coordinate sets; hence these derivatives are continuous at zero as well. This proves (b).

Suppose that the E-valued kth derivative at zero existed. Applying the continuous nth coordinate projection would give

$$\big(f_k^{(k)}(0)\big)_n = \frac{d^k}{dt^k}\big(t^k \chi(nt)\big)\Big|_{t=0} = k! \qquad (n \geq 1),$$

because χ is identically one near zero. The derivative would therefore have to be $k!(1, 1, \ldots)$, which is not an element of c_{00}. This contradiction proves (c). $\square$

4 Sharp context: the Banach-space case

The first sentence of the source question also asks whether there are infinite-dimensional targets beyond the semi-Montel class. For Banach spaces this was already answered by Bachir and Lancien [2].

Proposition 1 (Bachir–Lancien). *If X is a Banach space with the Schur property, U is open in a Banach space, and $f : U \to X$ is weakly $\mathcal{C}^k$ for a positive integer k, then $f \in \mathcal{C}^k(U, X)$.*

Thus ℓ^1 is an infinite-dimensional positive example. Their paper also characterizes the Schur property through weak differentiability. For completeness, the finite-order converse can be seen directly. If X is not Schur, choose a weakly null sequence (x_n) with $\|x_n\| = 1$. Put $a_n = 4^{-n}$, $\delta_n = a_n/8$, take a nonzero $\varphi \in \mathcal{C}_c^\infty((-1, 1))$, and paste the disjoint bumps

$$F(t) = a_n^k \varphi\left(\frac{t - a_n}{\delta_n}\right) x_n \quad \text{for } |t - a_n| < \delta_n, \qquad F(t) = 0 \quad \text{otherwise.}$$

All scalar kth derivatives tend to zero at the accumulating point because $x'(x_n) \to 0$ for every $x' \in X'$, whereas at a fixed nonzero derivative point of φ the norms of the vector kth derivatives stay bounded away from zero. Hence F is weakly $\mathcal{C}^k$ but not norm-$\mathcal{C}^k$. For each fixed finite $k \geq 1$, the Banach full weak–strong principle is therefore equivalent to the Schur property.

The new contribution of Theorem 1 is different: it settles the source's explicit proposal that *every* generalised Schwartz space might work. It shows that precompact bounded sets do not prevent a derivative from escaping into the completion.

5 Verification, novelty, and limitations

The accompanying finite-window script checks the support bound and the forced derivative prefixes for $1 \leq k \leq 8$. These computations are only sanity checks; the infinite-support contradiction is the proof.

A bounded web and literature search on 13 August 2026 used the exact source title together with the phrases "generalised Schwartz weak–strong," "c_{00} weak–strong differentiable," and "$\mathcal{C}^k$ weak–strong Schur." It recovered the source and Bachir–Lancien, but no occurrence of the counterexample above. The novelty verdict is therefore *apparently new within the bounded search*, not an exhaustive bibliographic guarantee.

The result answers Question 5.3.9(i)'s generalised-Schwartz follow-up for all positive finite orders. It does not settle parts (ii)–(iv) in their full locally convex generality. Human review should focus on the standard dual and precompactness facts in Lemma 1.

References

[1] K. Kruse, *On vector-valued functions and the ε-product*, arXiv:2301.13612 (2023), Question 5.3.9(i), p. 109.

[2] M. Bachir and G. Lancien, *On the composition of differentiable functions*, Canadian Mathematical Bulletin **46** (2003), 481–494, doi:10.4153/CMB-2003-047-2.